

Human Attention Is Not All You Need: From Slop-vestigations to Systematic Automated Review

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1 Introduction

In July 2026, while OpenAI was training a model to be highly persistent and collaborative, over 1200 agents across different models collectively hacked Hugging Face and attempted to deceive researchers that they had accomplished their tasks legitimately during evaluation (METR, 2026). Beyond being one of the most prominent cases of AI misalignment thus far, it emphasizes the need for robust, tractable oversight and auditing mechanisms for agent outputs and trajectories. Ryan Greenblatt, one of the three METR and Redwood Research staff members who were allowed access by OpenAI to incident-relevant logs over the course of six days, lamented their reliance on suboptimal analysis agents to investigate the incident due to the need to analyze “over a thousand extremely long transcripts from agents that ran for multiple days”. He called it a “slop-vestigation”, where the analysis agents were often overconfident in their explanations, missed important details, and failed to understand what happened (Greenblatt, 2026).

As agents become capable of accomplishing more complex tasks with longer time horizons, it will become increasingly difficult to detect instances of misalignment when they occur and in retrospect. In this project, we will explore automated procedures to audit agent trajectories that improve the efficacy and efficiency of human review.

2 Research Questions

Our key research questions are as follows:

1. How does analysis agent debate compare to human and vanilla agent baselines?
2. What changes in performance between different variants of debate, such as between

standard and prover-estimate debate?

3. How much does finetuning models for debate improve analysis performance?
4. What tradeoffs exist between minimizing flagging false positives and missing true negatives in different contexts?
5. Can we develop techniques for which we can assert useful performance guarantees?
6. How does the performance of the techniques we will analyze depend on the computational budget allocated to them?

3 Literature Review

In (Brown-Cohen et al., 2026) the authors define prover-estimator debate by making the prover split the claim into smaller claims, and subsequently the estimator gives a probability score for each claim. Once the debate is concluded, the judge will review a small final claim. In (Brown-Cohen et al., 2023) the authors present that in some debates the judge LLM checks only a few steps and can still reach the correct decision, even if the review LLM’s are more capable models than the judge LLM. (Lù et al., 2025) presents a dataset that has agent trajectories labeled by experts as success or failure. On this dataset, the authors have tested 12 AI judges, and they concluded that no AI judge is fully reliable on every benchmark. In our method, we try to narrow down the error to specific spots in the sequence of texts/commands so a human reviewer can focus on those particular steps. (Zhang et al., 2025) present the **Who&When** dataset, this dataset contains fine grained failure annotations obtained from 127 LLM’s across 184 distinct tasks. Each sample contains a query, a failure log, an agentic system configuration, and fine grained expert annotations.

We will be using this dataset as the main dataset for experimentation.

4 Tasks, Benchmarks, and Environments

The main task we consider is analyzing an agent trajectory log and drawing attention to critical steps in the trajectory for human review. We thus consider benchmarks including trajectories where experts have already labeled critical steps. We compare each method’s answer with this label.

Some datasets we could use are:

1. Who&When
2. MP-Bench
3. AgentProcessBench
4. Who&When
5. TraceElephant

Subject to time constraints, we may or may not use all of these datasets.

Our tentative plan is to first attempt to use Olama on Mac and then conduct the full experiment on Kaggle, where we will have access to 30 GPU hours per person per week. We will first use Qwen3.5-9B for the debater agents and Qwen3.5-2B, a smaller model, as the judge agents, later testing our techniques with other models. All numbers in the paper will come from Kaggle. If we are ahead of schedule, then we may need paid API for finetuning or running larger models.

5 Methodology

Each method receives one agent trajectory and identifies the critical steps. We compare this step with the critical step label.

1. Prover-Estimator Debate

The prover argues whether the trajectory is successful or unsuccessful, dividing the trajectory into sections and making sub-claims about each section. The estimator gives a probability for each sub-claim corresponding to its credence. The debate then recurses with each iteration, with the prover refining its argument based on the estimator’s credences on increasingly small sections of the trajectory.

2. Standard Debate

Two debaters argue about which steps are critical. They argue for N rounds, after

which a judge summarizes the critical steps based on the debate.

3. One-Shot

One AI model reads the full trajectory and selects the critical step. This is the simplest automated method.

4. Single Critique

One model proposes a step. A second model critiques it one time. Then a judge decides. In (Elasky et al., 2026) the results show that this simple method can be almost as effective as debate.

Each automated method will be allocated roughly the same number of tokens to ensure a fair comparison.

5. Human Baseline

The three authors find the critical steps without AI help.

5.1 Evaluation

We measure three things:

1. Are the steps correctly identified as critical? Each method selects the critical steps. We compare it with the critical step label.
2. How many computational resources do the methods use? We measure this in terms of time, token usage, and cost.
3. How much human time do the methods save? The three authors check approximately 30 cases and measure their time. Sometimes we read the full trajectory, and other times we read only the step from the debate. The difference gives the review minutes, and if needed we will review more.

6 Future Plan

Weeks 1–2: Start small: Prepare the project folder, download and take a look at the first dataset, make the simple pipeline run on local device, the main debate structure on a few trajectories, decide what to keep and change.

Weeks 3–4: Add the other methods: add the standard debate as well and the matched-call baselines, select a main test set.

Weeks 5–6: Move to Kaggle: Run all the methods on the dev dataset and improve the prompts, and decide if we also have capacity for optional task.

Weeks 7–8: Run the experiments: run all methods on test set, check about 30 trajectories by hand, and maybe prepare for optional task.

Weeks 9–10: Write: Write a nice paper with all findings so we can at least get on arxiv (<https://arxiv.org/>)

The code for this project can be found here (<https://github.com/shoimya/Human-Attention-Is-Not-All-You-Need>)

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